

A sharp covariance-only Talagrand deficit bound

Full solution packet for arXiv:2201.12478

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9 August 2026

Abstract

Bez, Nakamura, and Tsuji ask whether their sharp Talagrand deficit estimate for $0 < \beta < 1$ remains true under the sole hypothesis $\text{Cov}(\mu) \leq \beta \text{I}$. We prove this and a sharper anisotropic theorem. If $0 < \Sigma = \text{Cov}(\mu) \leq \text{I}$, then

$$\frac{1}{2}W_2(\gamma, \mu)^2 - H(\mu \mid \gamma) \leq \frac{1}{2} \log \det \Sigma - \text{tr} \sqrt{\Sigma} + n.$$

Equality holds exactly for translates of $N(0, \Sigma)$. The proof is a short combination of the Brenier change-of-variables identity, Gaussian integration by parts, concavity of $\log \det$, and a block-covariance inequality.

1 Source question

Let $\gamma = N(0, \text{I})$ on $\mathbb{R}^n$, and write

$$H(\mu \mid \gamma) = \int_{\mathbb{R}^n} \log \left(\frac{d\mu}{d\gamma} \right) d\mu.$$

The source paper [1] proves, under a β -semi-log-concavity hypothesis, that

$$\frac{1}{2}W_2(\gamma, \mu)^2 - H(\mu \mid \gamma) \leq n \left(1 + \frac{1}{2} \log \beta - \sqrt{\beta} \right) \quad (0 < \beta < 1). \quad (1)$$

In Section 1.4 it asks whether (1) remains true under only $\text{Cov}(\mu) \leq \beta \text{I}$. Figure 1 reproduces Theorem 1.5, the explicit sharp value, and the complete two-page question statement.

Here covariance is centered, so translations are permitted. In fact, every translate of $N(0, \beta \text{I})$ attains equality in (1).

2 Proof intuition

Let T be the Brenier map carrying the standard Gaussian to μ . The Talagrand deficit can be written using only the Jacobian $A = \nabla T$:

$$\frac{1}{2}W_2^2 - H = \mathbb{E}[\log \det A] - \text{tr}(\mathbb{E}A) + n.$$

Concavity of $\log \det$ says that replacing A by its mean can only increase this expression. Gaussian integration by parts identifies $M = \mathbb{E}A$ with the cross-covariance of $T(X)$ and X . Positivity of their joint covariance then gives $M^2 \leq \text{Cov}(\mu)$. The remaining scalar-looking matrix functional $\log \det M - \text{tr} M + n$ is increasing in Loewner order below the identity. This both proves the conjecture and retains the full covariance spectrum.

for the Hamilton–Jacobi equation, the dual form of Talagrand’s inequality, as well as the Poincaré and Beckner inequalities; see Section 6.

In our next result, we improve Mikulincer’s deficit estimate of Talagrand’s inequality (1.12) under somewhat stronger assumptions on the inputs.

Theorem 1.5. *Let $\beta > 0$. Suppose the twice differentiable $v : \mathbb{R}^n \rightarrow (0, \infty)$ satisfies $\int_{\mathbb{R}^n} v \, dx = 1$ and $\int_{\mathbb{R}^n} |x|^2 v \, dx < \infty$.*

(1) *If $\beta > 1$ and v satisfies*

$$0 \geq \nabla^2 \log v \geq -\frac{1}{\beta} \text{id},$$

then we have

$$(1.23) \quad \frac{1}{2} W_2(\gamma, v)^2 - \text{Ent}_\gamma\left(\frac{v}{\gamma}\right) \leq \frac{1}{2} W_2(\gamma, \gamma_\beta)^2 - \text{Ent}_\gamma\left(\frac{\gamma_\beta}{\gamma}\right).$$

(2) *If $0 < \beta < 1$ and v is β -semi-log-concave, then we have (1.23).*

A direct calculation shows that the constant in (1.23) is explicitly given by

$$\frac{1}{2} W_2(\gamma, \gamma_\beta)^2 - \text{Ent}_\gamma\left(\frac{\gamma_\beta}{\gamma}\right) = n \left(1 + \frac{1}{2} \log \beta - \sqrt{\beta}\right).$$

For any $\beta > 0$ we have

$$1 + \frac{1}{2} \log \beta - \sqrt{\beta} < -\frac{2(1 - \beta) + (\beta + 1) \log \beta}{2(\beta - 1)}$$

and so it is clear that (1.23) improves upon (1.12) for β -semi-log-concave inputs and $0 < \beta < 1$. As a further appealing feature, inequality (1.23) is clearly sharp since equality holds when $v = \gamma_\beta$.

We have already remarked that the standard argument for deriving Talagrand’s inequality from the LSI is not applicable in the setting of deficit estimates under a small-covariance assumption, and since the same appears to be true in our framework, we must prove Theorem 1.5 directly without appealing to Theorem 1.4 (or Theorem 1.3). Interestingly, our proofs of Theorems 1.3 and 1.5 are completely different; in the former case we use a flow monotonicity approach using Fokker–Planck flows, and in the latter case we use mass transportation. We refer the reader forward to Section 2.3 for further details of our methodology, as well as the preceding part of Section 2 in which we give a number of further extended remarks on our main results Theorems 1.3–1.5. Before that, we mention some related open problems which we consider to be interesting and were unable to address in the present work.

1.4. A few open problems. It seems reasonable to expect that (1.23) holds when $0 < \beta < 1$ and $\text{cov}(v) \leq \beta \text{id}$. If true, this would simultaneously extend both

Figure 1: Pages 10–11 of arXiv:2201.12478v3: inequality (1.23), its sharp Gaussian value, and the covariance-only open problem. The image retains the complete page width and stitches the two-line continuation from page 11.

3 Sharp anisotropic theorem

Theorem 1 (Covariance-sharp Talagrand deficit). *Let μ be an absolutely continuous probability measure on $\mathbb{R}^n$ with finite second moment, and let $\Sigma = \text{Cov}(\mu)$. If $0 < \Sigma \leq \text{I}$, then*

$$\boxed{\frac{1}{2}W_2(\gamma, \mu)^2 - H(\mu \mid \gamma) \leq \frac{1}{2} \log \det \Sigma - \text{tr} \sqrt{\Sigma} + n.} \quad (2)$$

If the relative entropy is finite, equality holds if and only if μ is a translate of $N(0, \Sigma)$.

In particular, if $0 < \beta < 1$ and $\Sigma \leq \beta \text{I}$, then

$$\frac{1}{2}W_2(\gamma, \mu)^2 - H(\mu \mid \gamma) \leq n \left(1 + \frac{1}{2} \log \beta - \sqrt{\beta} \right). \quad (3)$$

Equality in (3) holds if and only if μ is a translate of $N(0, \beta \text{I})$.

Proof. If $H(\mu \mid \gamma) = +\infty$, the assertion is immediate, so assume it is finite. We first give the argument in the smooth nondegenerate setting; the standard approximation step is recorded at the end.

Let $X \sim \gamma$ and let $T = \nabla \varphi$ be the Brenier map transporting γ to μ ; see, for example, [2, Chapter 10]. Put $A(x) = \nabla T(x) = \nabla^2 \varphi(x) > 0$. The change-of-variables formula gives

$$H(\mu \mid \gamma) = \frac{1}{2} \mathbb{E}[|T(X)|^2 - |X|^2] - \mathbb{E} \log \det A(X). \quad (4)$$

Since T is the optimal quadratic-cost map,

$$W_2(\gamma, \mu)^2 = \mathbb{E}|T(X) - X|^2. \quad (5)$$

Subtracting (4) from half of (5), and using $\mathbb{E}|X|^2 = n$, yields

$$\frac{1}{2}W_2^2 - H = n - \mathbb{E}[X \cdot T(X)] + \mathbb{E} \log \det A(X). \quad (6)$$

Let $m = \mathbb{E}T(X)$ and $M = \mathbb{E}A(X)$. Gaussian integration by parts gives

$$M = \mathbb{E}[(T(X) - m)X^\top], \quad \mathbb{E}[X \cdot T(X)] = \text{tr} M. \quad (7)$$

In particular, M is symmetric positive definite. Concavity of the matrix function $\log \det$ now turns (6) into

$$\frac{1}{2}W_2^2 - H \leq \log \det M - \text{tr} M + n =: F(M). \quad (8)$$

The joint covariance matrix of $(T(X), X)$ is

$$\begin{pmatrix} \Sigma & M \\ M & \text{I} \end{pmatrix} \geq 0.$$

Its Schur complement gives $M^2 \leq \Sigma$. By operator monotonicity of the positive square root,

$$0 < M \leq \sqrt{\Sigma} \leq \text{I}. \quad (9)$$

It remains only to observe that F is increasing in Loewner order on the positive matrices below I . Indeed, if $0 < B \leq C \leq \text{I}$ and $C_t = B + t(C - B)$, then

$$\frac{d}{dt} F(C_t) = \text{tr}[(C_t^{-1} - \text{I})(C - B)] \geq 0.$$

Applying this with (9) gives

$$\frac{1}{2}W_2^2 - H \leq F(\sqrt{\Sigma}) = \frac{1}{2} \log \det \Sigma - \operatorname{tr} \sqrt{\Sigma} + n,$$

which proves (2). If $\Sigma \leq \beta I < I$, the same monotonicity gives

$$F(\sqrt{\Sigma}) \leq F(\sqrt{\beta I}) = n \left(1 + \frac{1}{2} \log \beta - \sqrt{\beta} \right),$$

proving the source conjecture.

For equality, strict concavity of $\log \det$ in (8) forces $A(x)$ to be constant almost everywhere. Hence $T(x) = Mx + a$ and μ is a Gaussian translate. The covariance and the equality in (9) force $M = \sqrt{\Sigma}$. Equality in the scalar step further forces $\Sigma = \beta I$. Conversely, direct substitution shows that precisely these Gaussian translates attain equality.

For completeness, the same chain works directly without smoothness. Let A be the density of the absolutely continuous part of the distributional Hessian of φ , and define the cross-covariance $M = \mathbb{E}[(T(X) - m)X^\top]$. The generalized Monge–Ampère formula gives (4) with this Alexandrov derivative. Distributional Gaussian integration by parts gives

$$\mathbb{E}A \leq M,$$

because the singular part of the Hessian is a positive-semidefinite matrix-valued measure. Therefore Jensen and monotonicity of $\log \det$ give

$$\mathbb{E} \log \det A \leq \log \det(\mathbb{E}A) \leq \log \det M.$$

The block-covariance argument for M is unchanged, proving the stated generality. Equivalently, this passage can be obtained by the standard smooth approximation in the Brenier change-of-variables method [2, Chapter 10]. $\square$

4 Checks, scope, and novelty status

For $\mu = N(a, \Sigma)$, the standard Gaussian formulas give

$$H(\mu \mid \gamma) = \frac{1}{2}(\operatorname{tr} \Sigma + |a|^2 - n - \log \det \Sigma),$$

$$W_2(\gamma, \mu)^2 = |a|^2 + \operatorname{tr}(I + \Sigma - 2\sqrt{\Sigma}).$$

Their deficit is exactly the right side of (2); the mean cancels, as required. The accompanying script checks this identity and the scalar specialization on 10,000 random covariance spectra and translations.

The theorem resolves only the first problem in Section 1.4. It does not address the source’s separate semi-log-superharmonic hypercontractivity/LSI question or the appended Mahler discussion.

For novelty, the run indexes were searched by arXiv id, title, authors, and core phrases. A bounded primary-source web search through 9 August 2026 checked arXiv:2201.12478, Mikulincer’s small-covariance result arXiv:1906.05904, the dimensional Talagrand refinements in arXiv:1507.01086, and later searches combining “Talagrand deficit”, “small covariance”, “Brenier”, and “log det”. No explicit resolution of the question or statement of Theorem 1 was found. This is bounded evidence for candidate novelty, not an exhaustive claim.

5 Human-review recommendation

The core proof is the four-step chain

$$\text{transport identity} \implies \text{Jensen} \implies M^2 \leq \Sigma \implies \text{Loewner monotonicity}.$$

The result appears likely valid and resolves the source conjecture sharply. A human reviewer should focus on the approximation paragraph and perform an independent literature search for the same argument under a covariance-matrix or Gaussian optimal-transport formulation.

References

- [1] N. Bez, S. Nakamura, and H. Tsuji, *Stability of hypercontractivity, the logarithmic Sobolev inequality, and Talagrand's cost inequality*, arXiv:2201.12478v3 (2022), especially Theorem 1.5 and Section 1.4.
- [2] C. Villani, *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften 338, Springer, 2009.